A portfolio optimization model using Genetic Network Programming with Control nodes

Yan Chen(陳豔)[a], Shingo Mabu(間普真吾)[a], Etsushi Ohkawa(大川悦史), Kaoru Shimada(嶋田香)[b], Kotaro Hirasawa(平澤宏太郎)[a]

[a] Graduated school of Information, Production and Systems, Waseda University, 2-7, Hibikino, Wakamatsu-ku, Kitakyushu, Fukuoka, Japan

[b] Information, Production and Systems Research Center, Waseda University, 2-7, Hibikino, Wakamatsu-ku, Kitakyushu, Fukuoka, Japan

Expert Systems with Application 36 (2009), p.p. 10735-10745

Presenter: Hung-Hsi Chen
Date: Oct. 28, 2009

Abstract

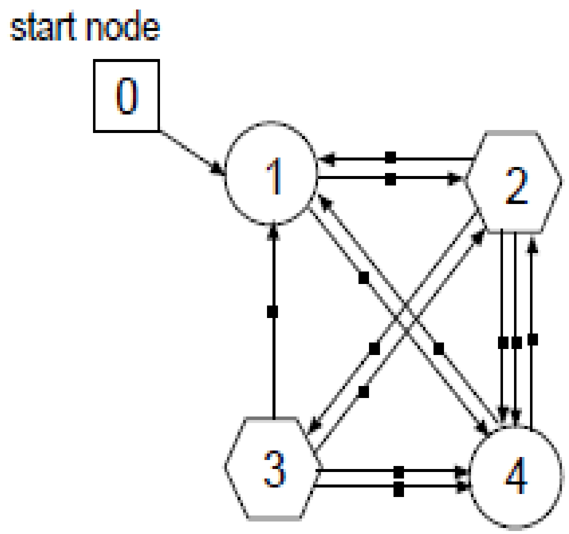
Many evolutionary computation methods applied to the financial field have been reported. A new evolutionary method named ”Genetic Network Programming” (GNP) has been developed and applied to the stock market recently. The efficient trading rules created by GNP has been confirmed in our previous research. In this paper a multi-brands portfolio optimization model based on Genetic Network Programming with control nodes is presented. This method makes use of the information from technical indices and candlestick chart. The proposed optimization model, consisting of technical analysis rules, are trained to generate trading advice.

Abstract

The experimental results on the Japanese stock market show that the proposed optimization system using GNP with control nodes method outperforms other traditional models in terms of both accuracy and efficiency. We also compared the experimental results of the proposed model with the conventional GNP based methods, GA and Buy&Hold method to confirm its effectiveness, and it is clarified that the proposed trading model can obtain much higher profits than these methods.

GNP-RL Structure

Directed graph structure



Gene structure

	Node gene			Connection gene			
node 0	0	—	—	1 0	—	—	—
node 1	2	1 0.3 5	2 0.0 5	2 0	—	4 0	—
node 2	1	2 2.0 1	1 0.0 1	3 0	4 0	1 0	4 0
node 3	1	1 0.0 1	3 0.1 1	4 0	1 0	2 0	4 0
node 4	2	4 0.7 5	1 2.7 5	2 0	—	1 0	—

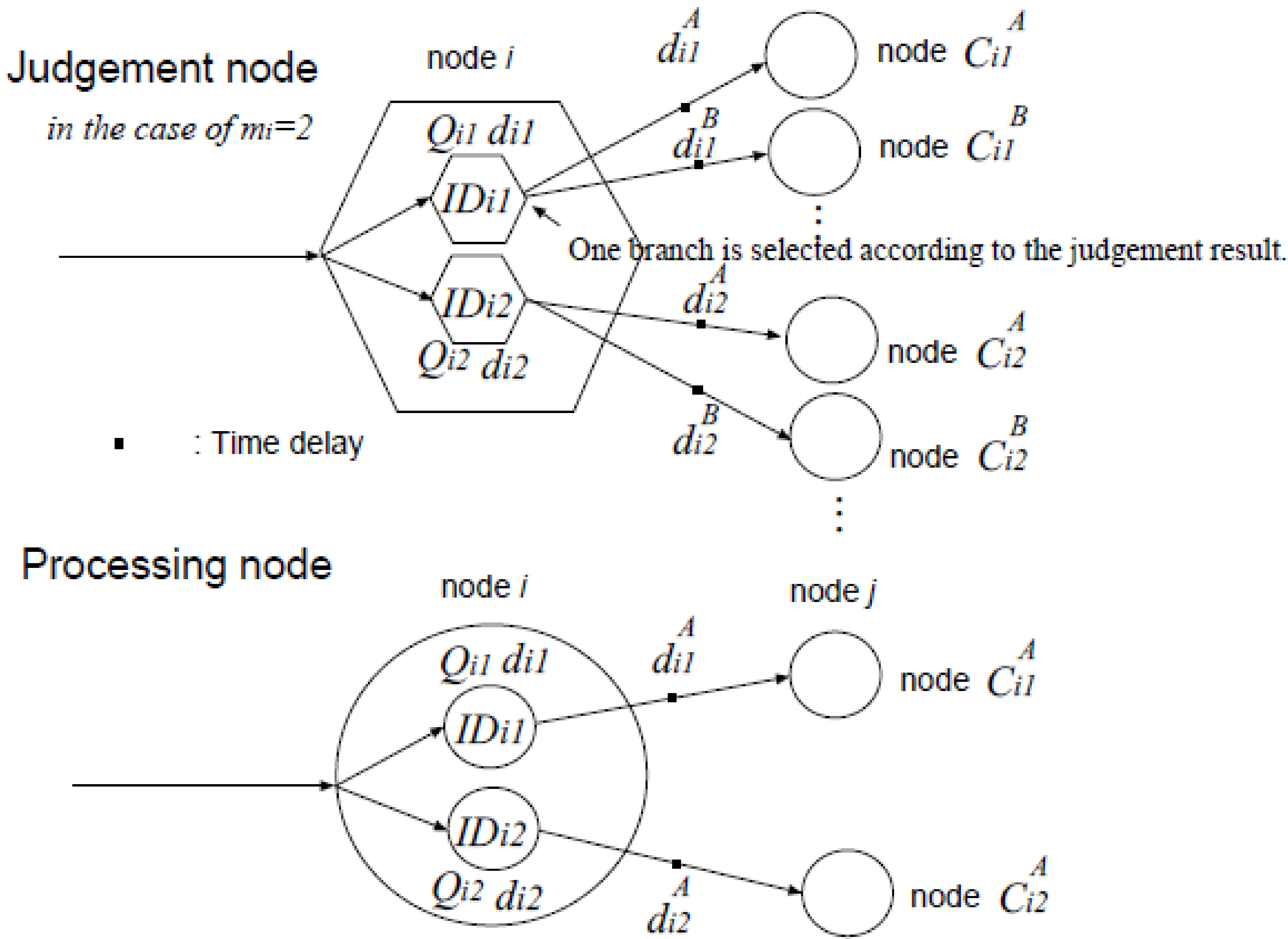
Node gene

node *i*	K_i	ID_{i1} Q_{i1} d_{i1}	.....	ID_{im_i} Q_{im_i} d_{im_i}

Connection gene

C_{i1}^A d_{i1}^A	C_{i1}^B d_{i1}^B	.....	.....	$C_{im_i}^A$ $d_{im_i}^A$	$C_{im_i}^B$ $d_{im_i}^B$

: Judgement node : Processing node : Time delay



Judgement node
in the case of mi=2
node i
Qi1 di1
IDi1
IDi2
Qi2 di2
One branch is selected according to the judgement result.
node Ci1 A
node Ci1 B
node Ci2 A
node Ci2 B
di1 A
di1 B
di2 A
di2 B
: Time delay
Processing node
node i
node j
Qi1 di1
IDi1
IDi2
Qi2 di2
di1 A
di2 A
node Ci1 A
node Ci2 A

A run of GNP-RL

If node i is judgment node, a maximum Q value among Q_{i1}~Q_{imi} is selected with the probability 1-ε, or a random one is selected with the probability of ε(ε-greedy policy)

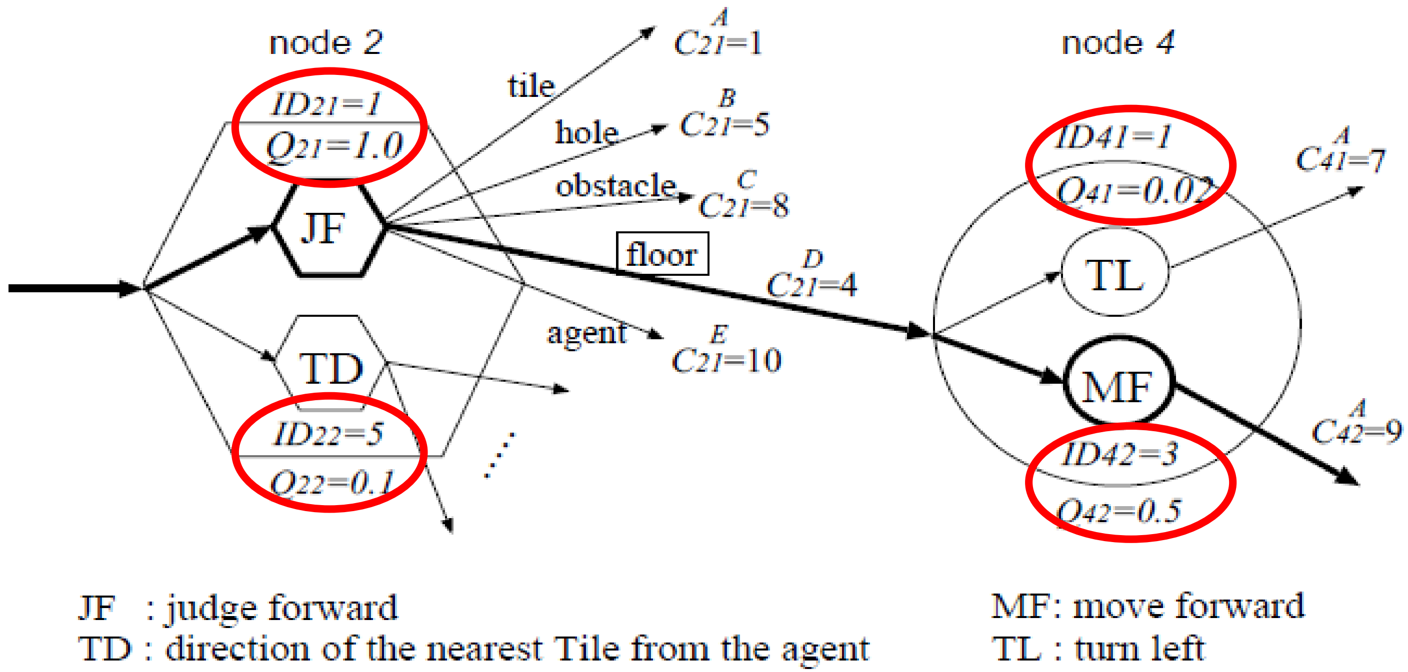


Figure 6: An example of node transition using the nodes for Tileworld problem.

GNP with control nodes(GNPcn)

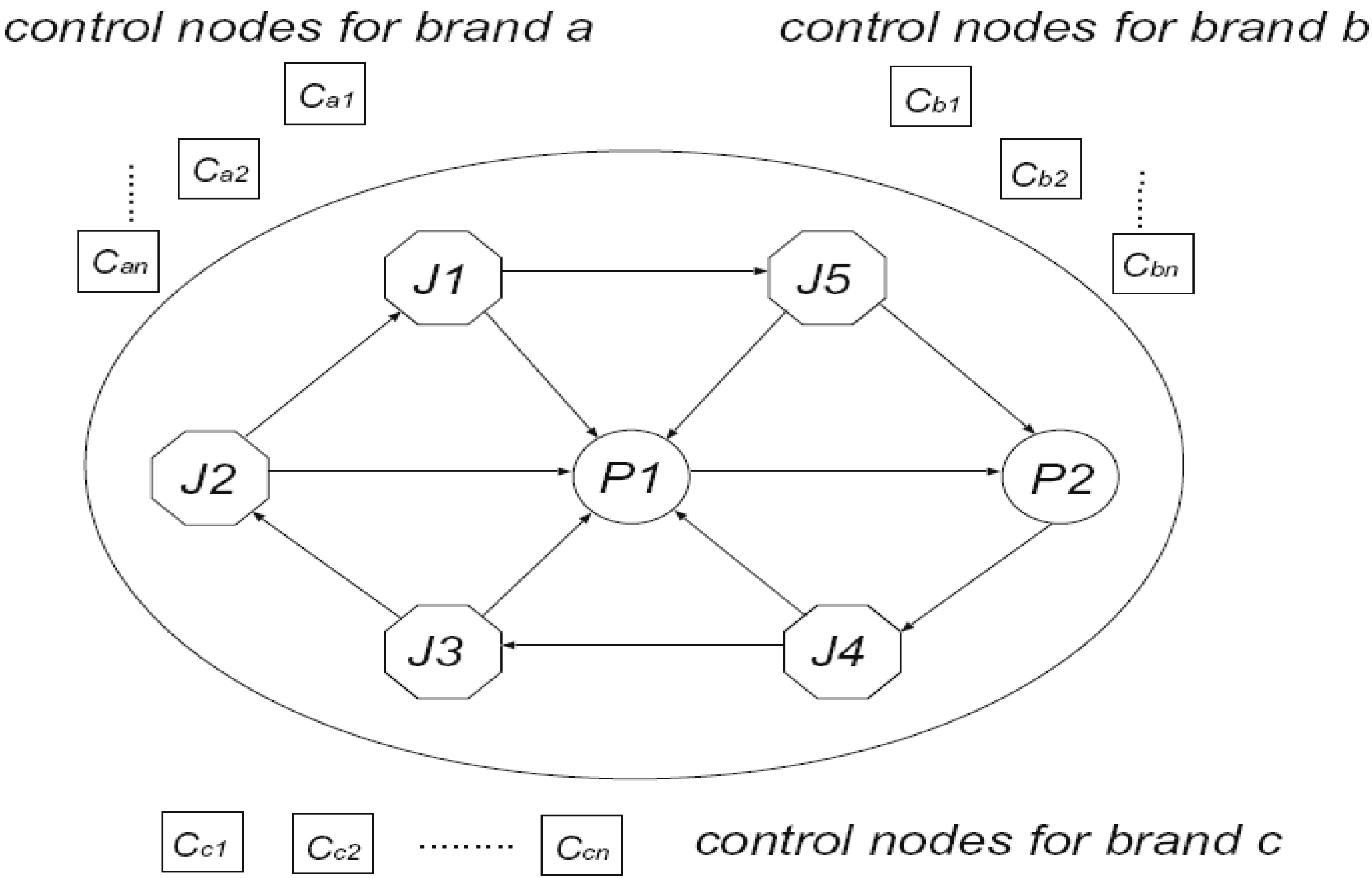
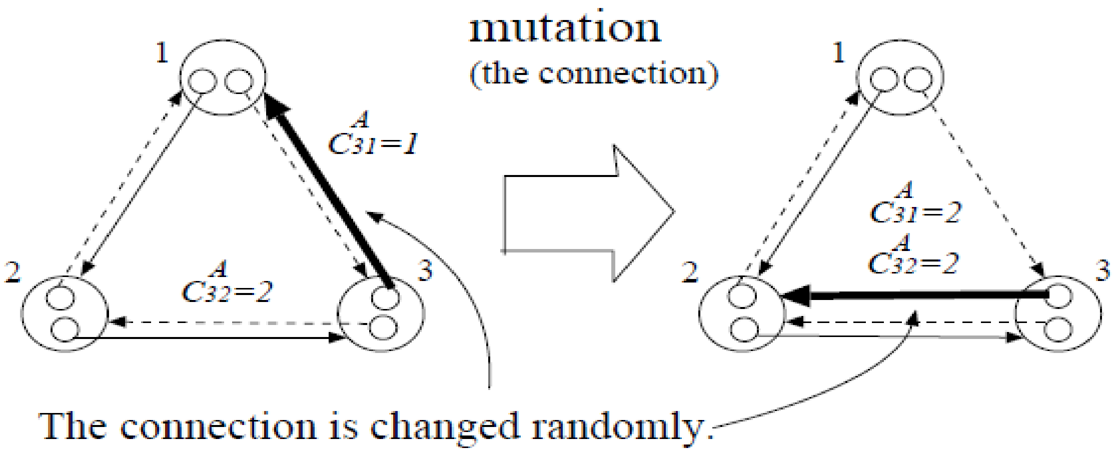


Fig. 2. The basic structure of GNPcn.

Mutation

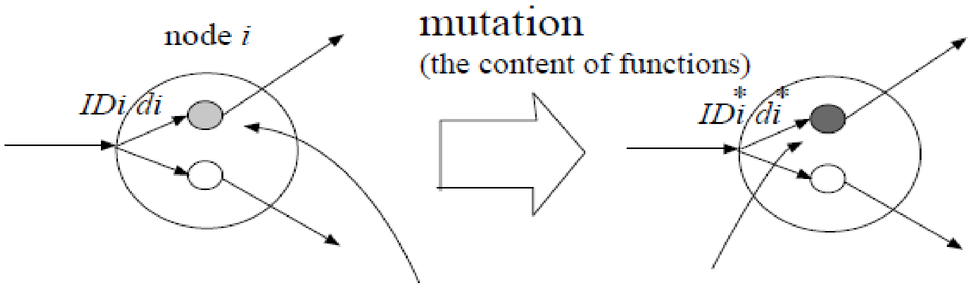
1、 Connection of functions: each node connection is re-connected to another node with the probability of Pm



node1	2	3 7.3 5	5 2.5 5	20	30
node2	2	2 3.2 5	4 2.3 5	30	10
node3	2	1 1.7 5	3 3.1 5	**10**	20

node1	2	3 7.3 5	5 2.5 5	20	30
node2	2	2 3.2 5	4 2.3 5	30	10
node3	2	1 1.7 5	3 3.1 5	**20**	20

2、 content of functions: each function is selected with the probability of Pm and changed to another function, i.e. ID_{ip} and d_{ip} are each changed.



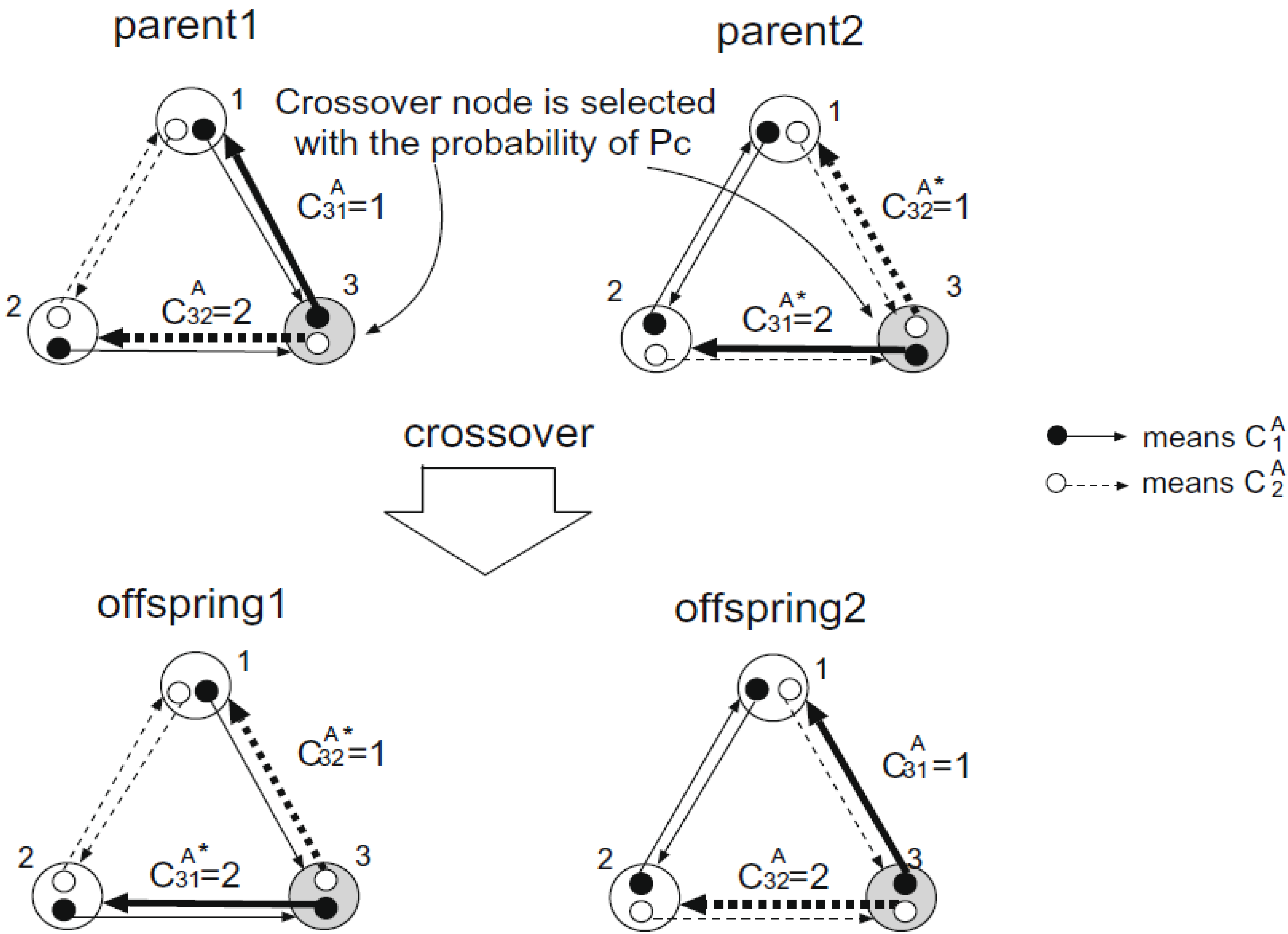
The content of the functions is changed randomly.

nodeX	2	3 7.3 5	5 2.5 5	20	30

nodeX	2	1 7.3 5	5 2.5 5	20	30

3、 Change node number: the number of processing nodes activated per control node(m) is changed to the other value with the probability of P_m.

Crossover



parent1
parent2
Crossover node is selected
with the probability of Pc
1
2
3
$C_{31}^{A}=1$
$C_{32}^{A}=2$
$C_{32}^{A*}=1$
$C_{31}^{A*}=2$
crossover
means C_{1}^{A}
means C_{2}^{A}
offspring1
offspring2
$C_{32}^{A*}=1$
$C_{31}^{A*}=2$
$C_{31}^{A}=1$
$C_{32}^{A}=2$

Learning phrase of GNPcn

RL is carried out when agents are carrying out their tasks and terminates when the time step reaches the predefined step.

Q(St, at) is updated as follow.(Sarsa algorithm):

Initialize $Q(s,a)$ arbitrarily
Repeat (for each episode):
    Initialize s
    Choose a from s using policy derived from Q (e.g., ε-greedy)
    Repeat (for each step of episode):
        Take action a, observe r, s'
        Choose a' from s' using policy derived from Q (e.g., ε-greedy)
        $Q(s,a) \leftarrow Q(s,a) + \alpha\left[r + \gamma Q(s',a') - Q(s,a)\right]$
        $s \leftarrow s'$; $a \leftarrow a'$;
    until s is terminal

Figure 6.9: Sarsa: An on-policy TD control algorithm.

α: step size parameter, γ: discount rate

Taken from the textbook: Reinforcement-An Introduction, Sutton, R.S. and Barto, A.G.

IMX functions

- If judgment of ROD are selected.
- The value of ROC is more than 0.1, the result becomes E.
- The next node number become C^{E}_{i1}.
- The shape of IMX functions is fixed here.

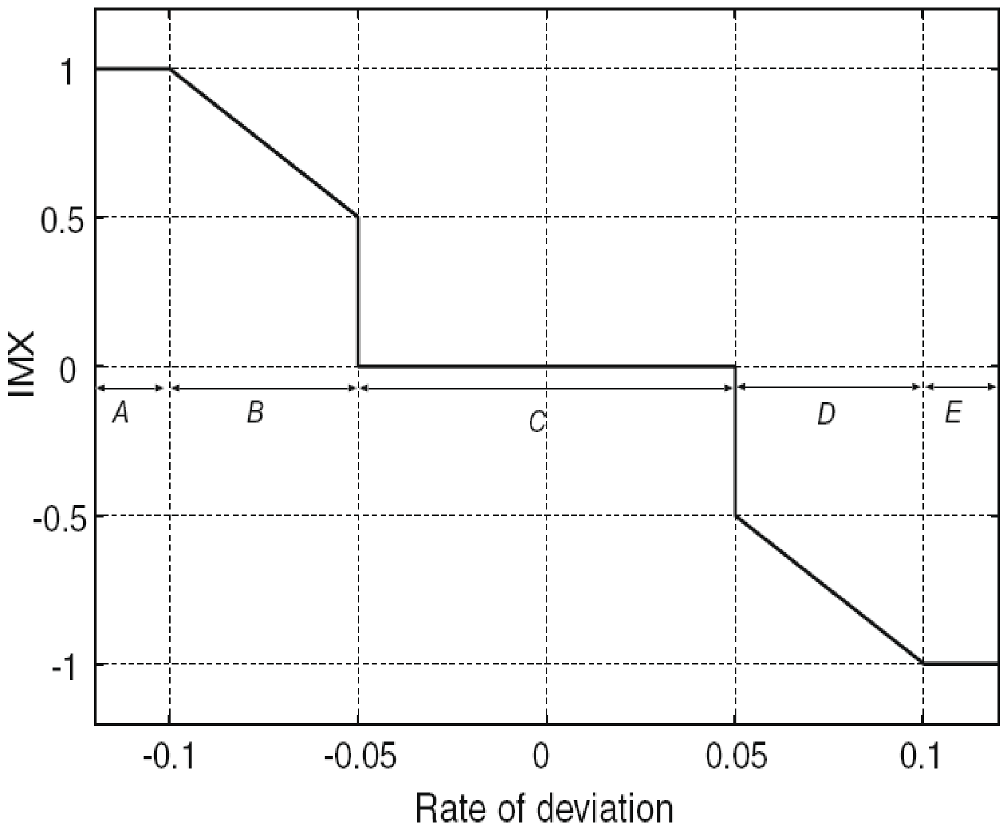


Fig. 5. IMX functions in judgment nodes (in case of ROD).

Technical indices:

ROD_m: $(C(t)-MA_m(t))/MA_m(t)$

RSI = 100 - (100/(1+RS))

RCI

VR= (Qu_n + Qf_n / 2) / (Qd_n + Qf_r

STC: position between %K(Quick) and %D(Slow)

ROC_n=(today's close – close n period ago)/(close n period ago)*100

Golden/Dead Cross:

- 3 days after a golden cross appears, IMX = 1
- 3 days after a dead cross appears, IMX = -1
- otherwise IMX = 0

MACD :

- 3 days after MACD passes through the signal from the lower side to the upper side, IMX = 1
- 3 days after it does from the upper to lower, IMX = -1
- Otherwise IMX = 0

Calculation periods of the technical index (day).

Technical index	Period1	Period2	Period3
Rate of deviation	6	12	26
RSI	6	12	26
ROC	6	12	26
Volume ratio	6	12	26
RCI	6	12	26
Stochastics	6	12	26
Golden/Dead cross	12 (short term)	26 (long term)	
MACD	12 (short term)	26 (long term)	9 (signal)

21 judgment functions: 18 + Gold/ Dead Cross + MACD + Candlestick

2 processing functions: buying and selling

Candlestick chart

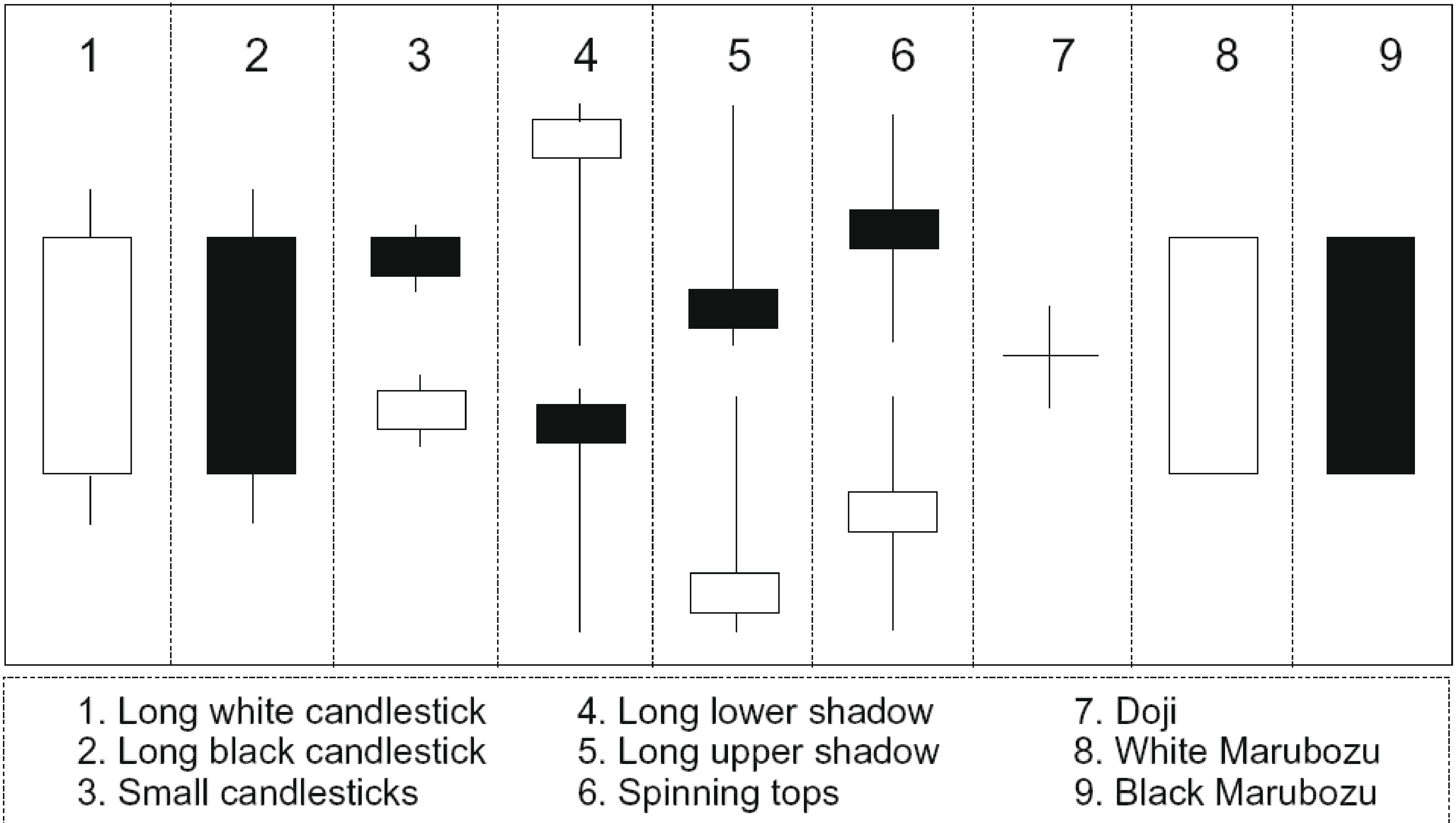


Fig. 4. Candlestick chart patterns.

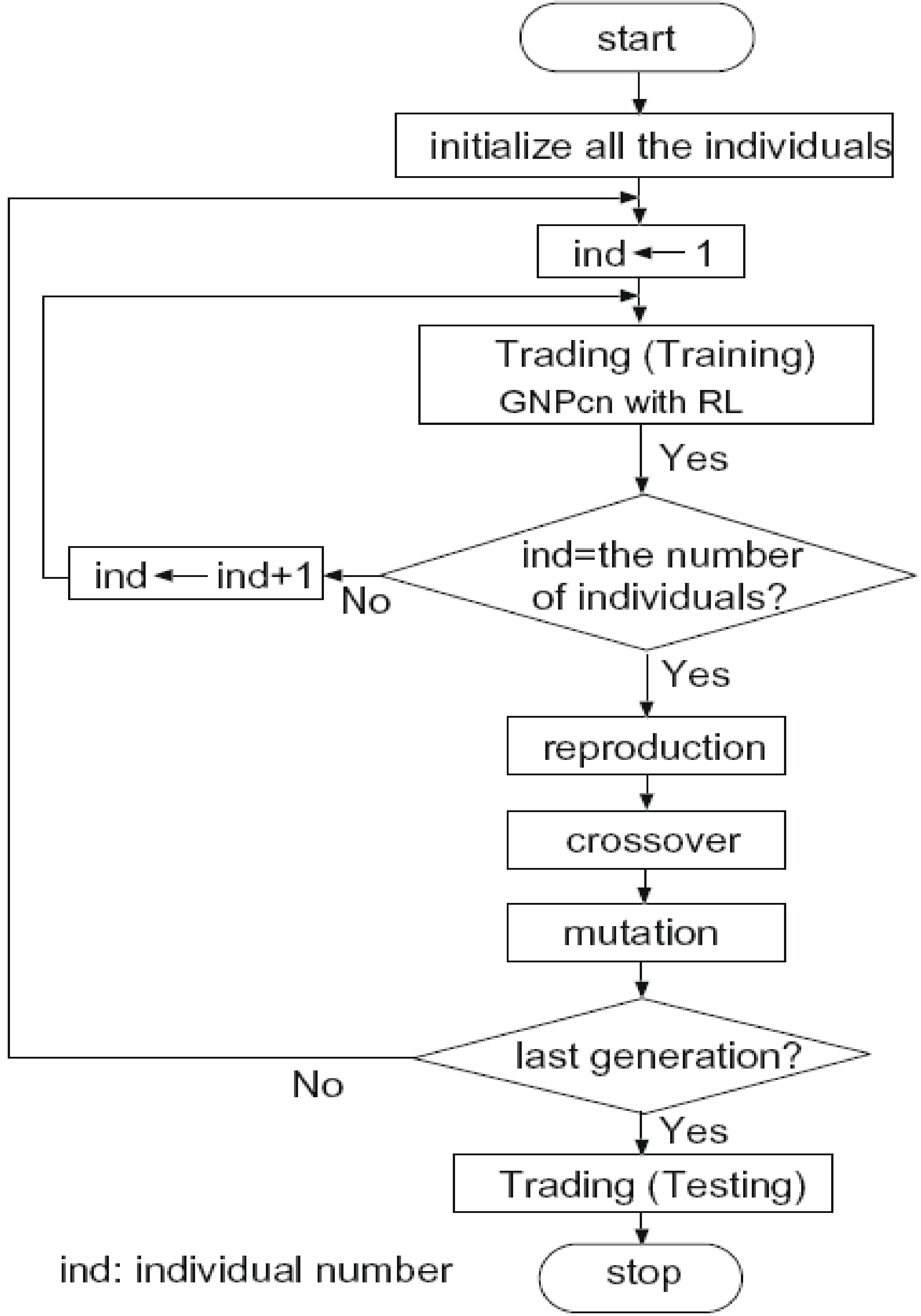


Fig. 3. Flowchart of GNPcn.

Notation

- $Initial(t)$: initial budget for training.
- $Initial(v)$: initial budget for validation.
- $Initial(t, b, n)$: initial budget of brand b in the nth generation for training.
- $Initial(v, b)$: initial budget of brand b for validation.
- T: temperature parameter.
- B: set of brands.
- b: brand index.
- N: number of generations.
- n: generation number.
- $Sell(t, b, n)$: sum of the money obtained by selling the stocks of brand b during the dealing in the nth generation for training.
- $Buy(t, b, n)$: sum of the money paid by buying the stocks of brand b during the dealing in the nth generation for training.
- $Profit(t, b, n)$: money gained or lost during the dealing of brand b in the nth generation for training, where $Profit(t, b, n) = Sell(t, b, n) - Buy(t, b, n)$.
- $Profitability(t, b, n)$: profitability during the dealing of brand b in the nth generation for training, where $Profitability(t, b, n) = Profit(t, b, n)/Initial(t, b, n)$.
- $Profit(v, b, d)$: money gained or lost during the dealing of brand b until day d for validation.

Training phase

$$Fitness(n) = \sum_{b \in B} Profit(t, b, n)$$

1st generation:

$$Initial(t, b, 1) = \frac{Initial(t)}{|B|}$$

$$Profit(t, b, 1) = Sell(t, b, 1) - Buy(t, b, 1)$$

$$Profitability(t, b, 1) = \frac{Profit(t, b, 1)}{Initial(t, b, 1)}$$

nth generation, n≠1:

$$Initial(t, b, n+1) = \frac{exp(Profitability(t, b, n)/T)}{\sum_{b \in B} exp(Profitability(t, b, n)/T)} Initial(t)$$

$$Profit(t, b, n) = Sell(t, b, n) - Buy(t, b, n)$$

$$Profitability(t, b, n) = \frac{Profit(t, b, n)}{Initial(t, b, n)}$$

Validation phase

$$Initial(v,b) = \frac{exp(Profitability(t,b,N)/T)}{\sum_{b \in B} exp(Profitability(t,b,N)/T)} Initial(v)$$

•At processing node, the trading is using the opening price.

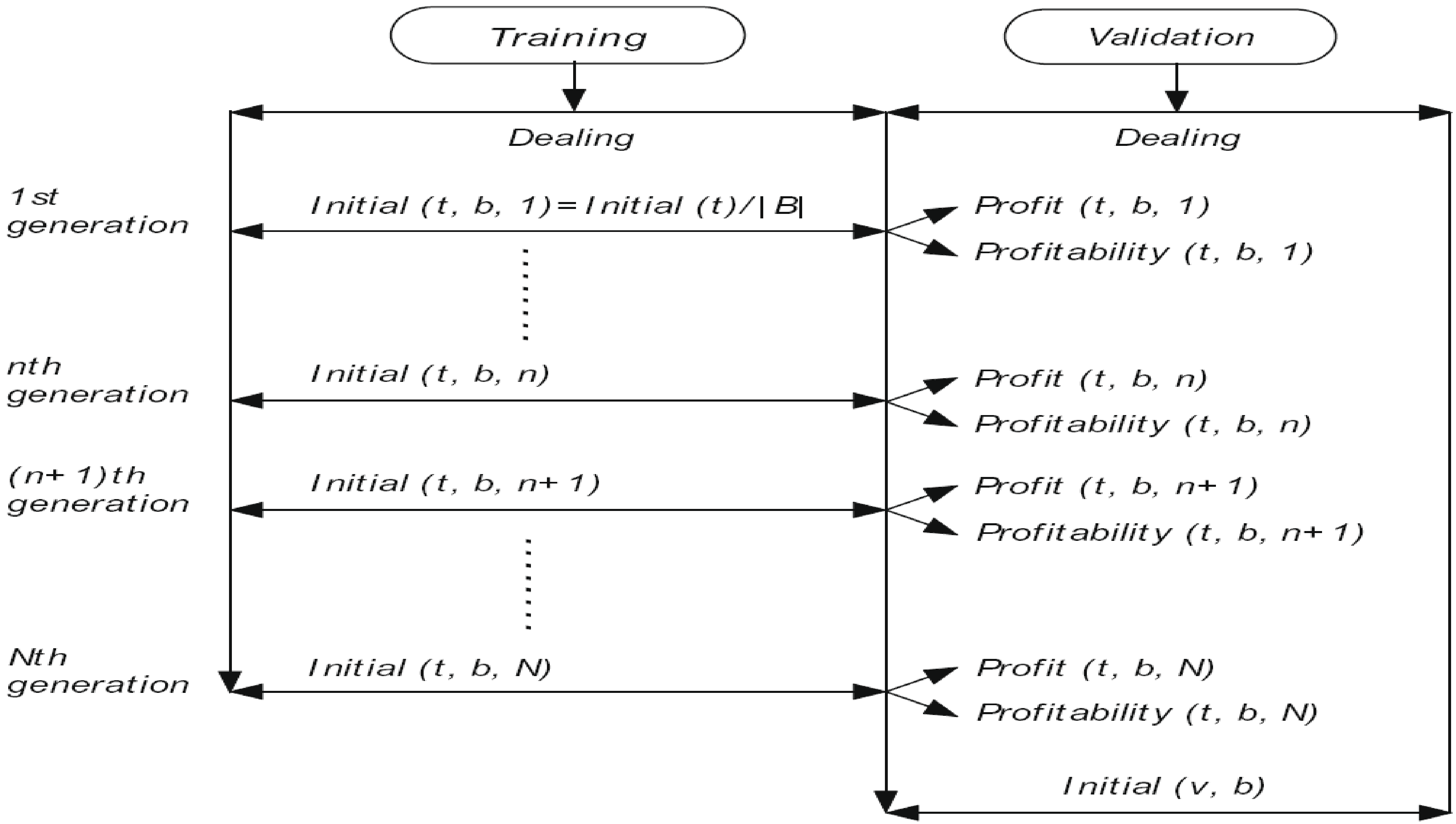


Fig. 6. Training and validation phase.

Experiment

- 10 stocks selected from Tokyo stock market in Japan
- Training: 01/04/2001 ~ 12/30/2003 (737 days)
- Testing: 01/05/2004 ~ 12/30/2004 (246 days)
- Initial funds = 5,000,000 (Japanese yen) in both period
- Buy and hold with the opening price

Table 3
Parameter conditions for evolving GNP.

Number of individuals = 300 (mutation:179, crossover:120, elite:1)
Number of nodes = 80 (judgement node = 20, Processing node = 10, control node = 50)
Number of sub-node in each node = 2
$P_c = 0.1$, $P_m = 0.03$, $\alpha = 0.1$, $\gamma = 0.3$, $\epsilon = 0.1$

Table 4 30 tests in the training.
Average fitness values [million yen] in training period with different *T*.

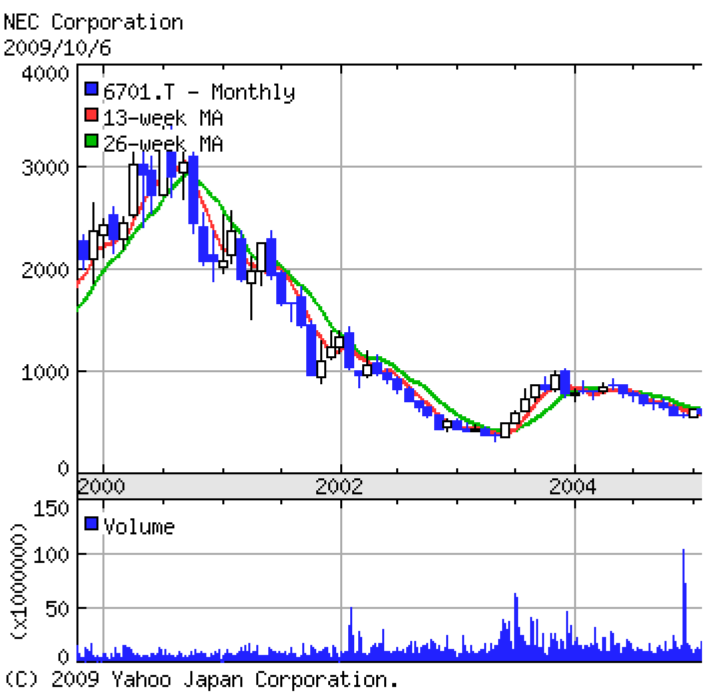
T	0.001	0.003	0.005	0.01	0.03	0.05
Fitness	133.5	$\boxed{143.2}$	136.2	125.6	110.1	108.2

- According to the Boltzmann distribution theory, different T value defines different strength distribution (the distribution of initial budget).
- If T->∞, the initial budget to each stock evenly.
- If T->0, all the money is distributed to the stock brand which obtain the best profitability.

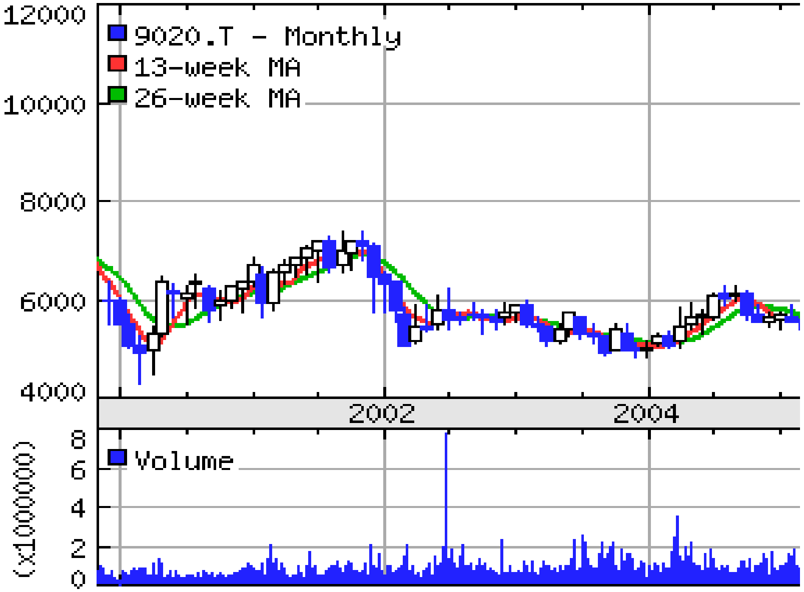
Table 2

Profits in the testing simulations {Profit[yen](profitability[%])}.

Brand	GNPcn	GNP-RL	GNP-candlestick	GA	Buy&Hold
NEC Corporation	–	43,400(0.9)	−126,150(−2.5)	−531,000(−10.6)	−1,026,000(−20.5)
Fuji Heavy Ind.	–	188,800(3.8)	97,700(2.0)	173, 000(3.5)	−189,000(−3.8)
East Japan Railway	–	418,800(8.4)	96,550(1.9)	7,590(0.2)	477,000(9.5)
KDDI	–	316,233(6.3)	−76,600(−1.5)	−273,000(−5.5)	−576,000(−11.5)
Nomura Holdings, Inc.	–	638,933(12.8)	−293,785(−5.9)	−374,087(−7.5)	−985,500(−19.7)
Shin-Etsu Chemical	–	133,800(2.7)	7,250(0.1)	−539,400(−10.8)	−264,000(−5.3)
Sony	-	202,766(4.1)	280,500(5.6)	112,000(2.2)	150,000(3.0)
Tokyo Electric Power	–	116,100(2.3)	−65,500(−1.3)	360,750(7.2)	262,500(5.3)
Hitachi	–	403,366(8.1)	472,300(9.4)	276,300(5.5)	336,000(6.7)
Nissan	–	464,966(9.3)	590,750(11.8)	372,000(7.4)	450,000(9.0)
Average	$\boxed{4,262,714(8.5)}$	292,716 (5.9)	98,302(2.0)	−41,585 (−0.8)	−136,500 (−2.7)

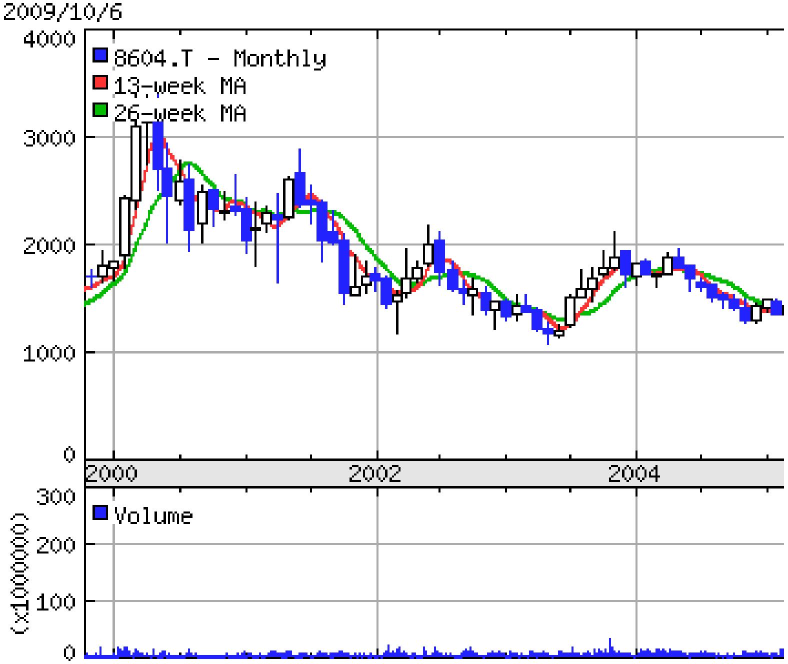


East Japan Railway Company
2009/10/27



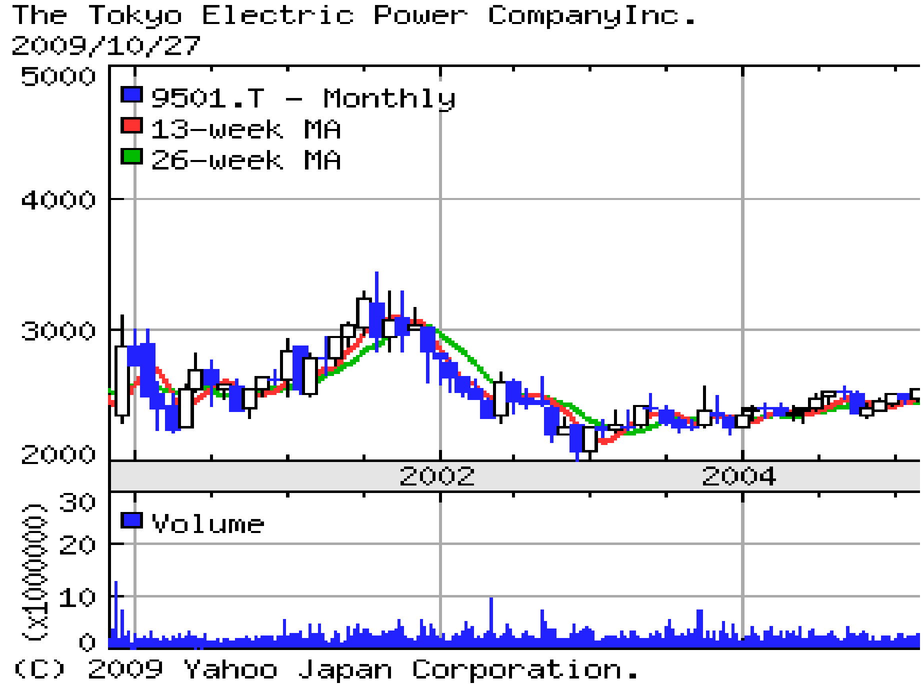
KDDI CORPORATION
2009/10/6

Nomura Holdings Inc.
2009/10/6

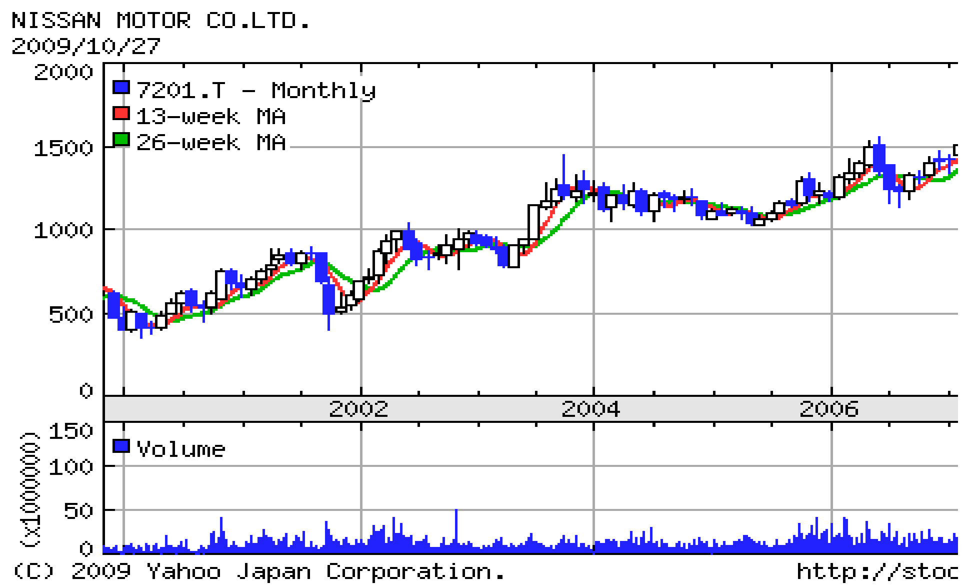


Shin-Etsu Chemical Co.Ltd.
2009/10/6


SONY CORPORATION
2009/10/6
6758.T - Monthly
13-week MA
26-week MA
20000
15000
10000
5000
0
2000
2002
2004
Volume
60
40
20
0
(x1000000)
(C) 2009 Yahoo Japan Corporation.


The Tokyo Electric Power CompanyInc.
2009/10/27
9501.T - Monthly
13-week MA
26-week MA
5000
4000
3000
2000
2002
2004
Volume
30
20
10
0
(x1000000)
(C) 2009 Yahoo Japan Corporation.


Hitachi Ltd.
2009/10/27
6501.T - Monthly
13-week MA
26-week MA
2000
1500
1000
500
0
2002
2004
Volume
150
100
50
0
(x1000000)
(C) 2009 Yahoo Japan Corporation.


NISSAN MOTOR CO.LTD.
2009/10/27
7201.T - Monthly
13-week MA
26-week MA
2000
1500
1000
500
0
2002
2004
2006
Volume
150
100
50
0
(x1000000)
(C) 2009 Yahoo Japan Corporation.
http://stoc

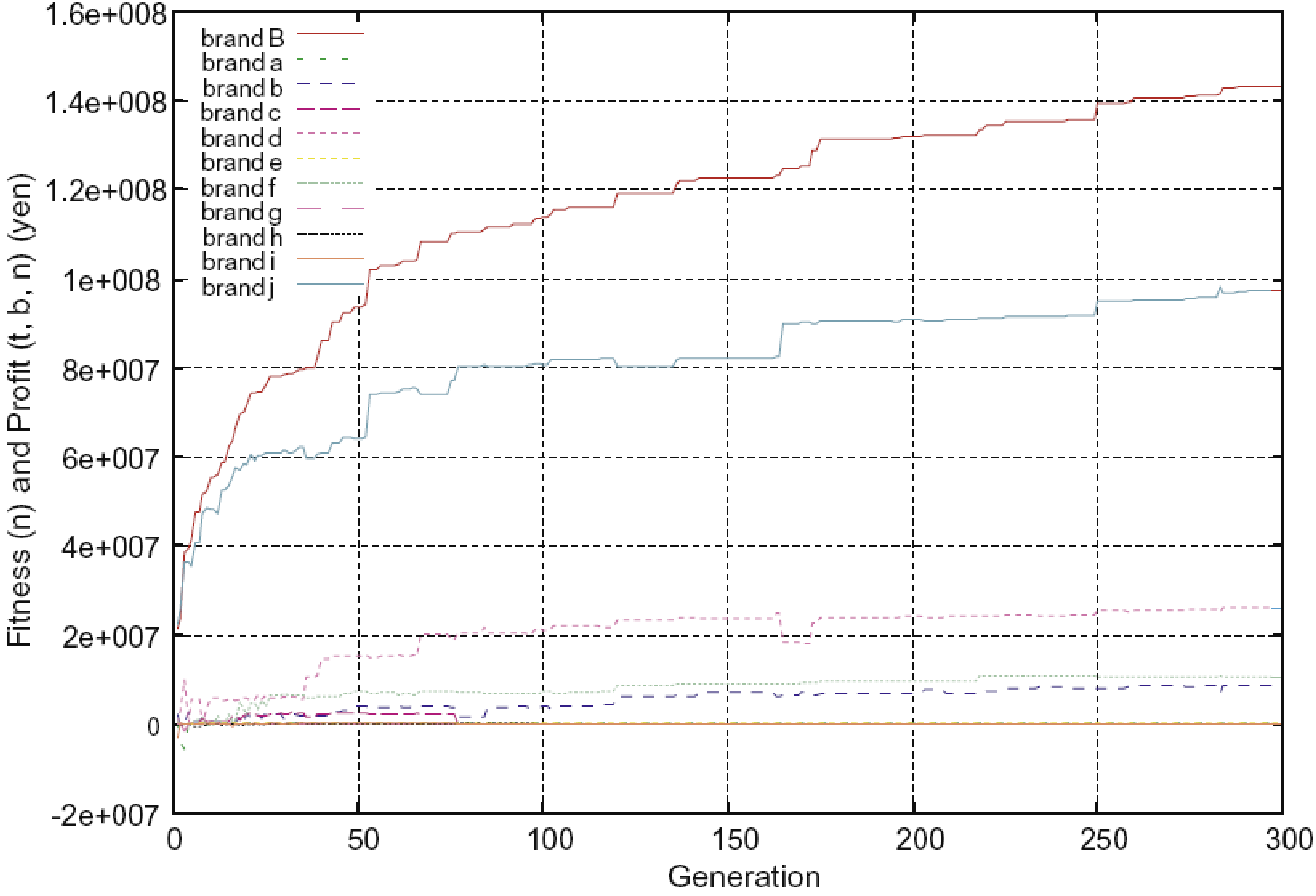


Fig. 7. Fitness and profit curves of 10 brands in the training period.

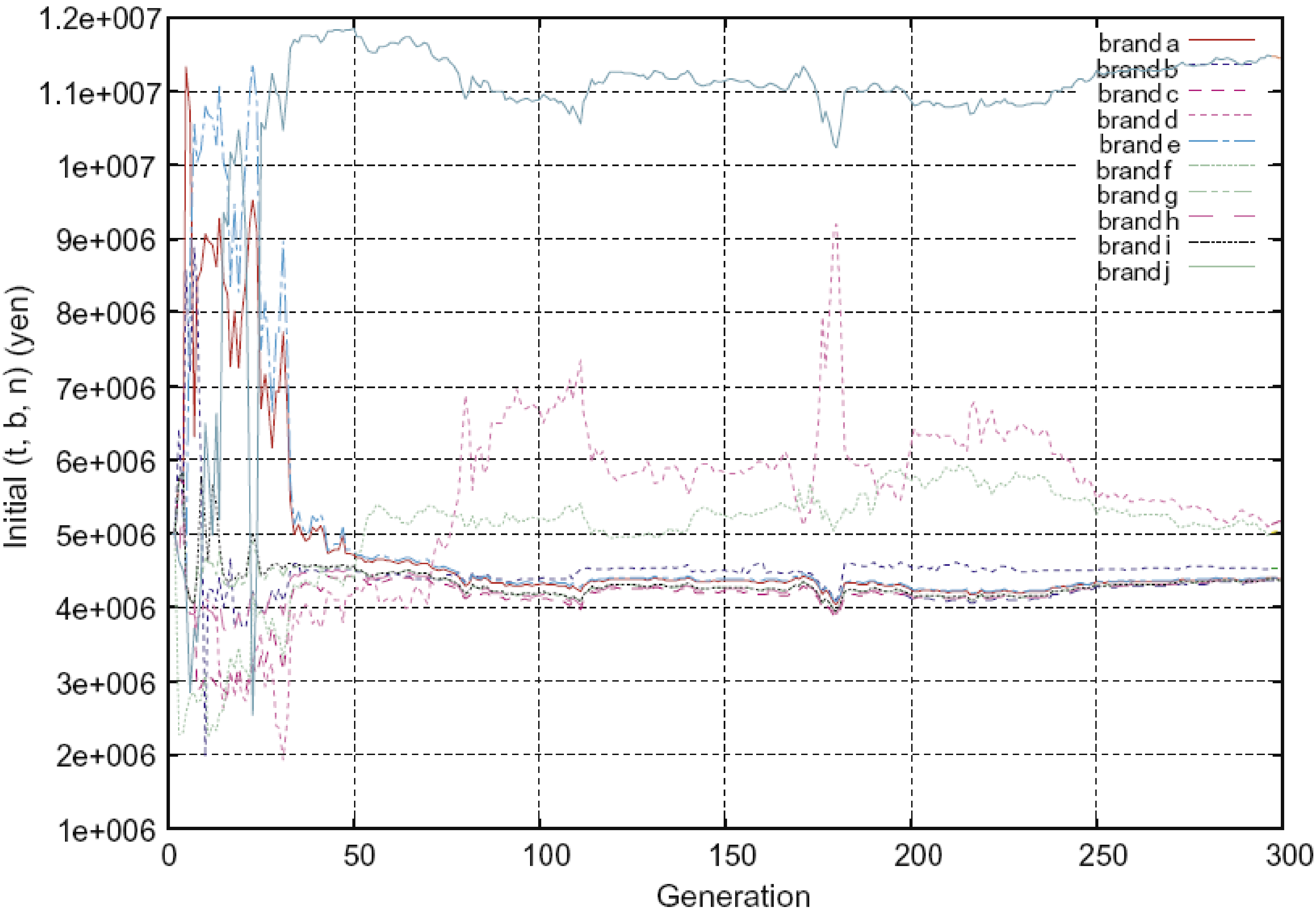


Fig. 8. Initial budget change of 10 brands in the training period.

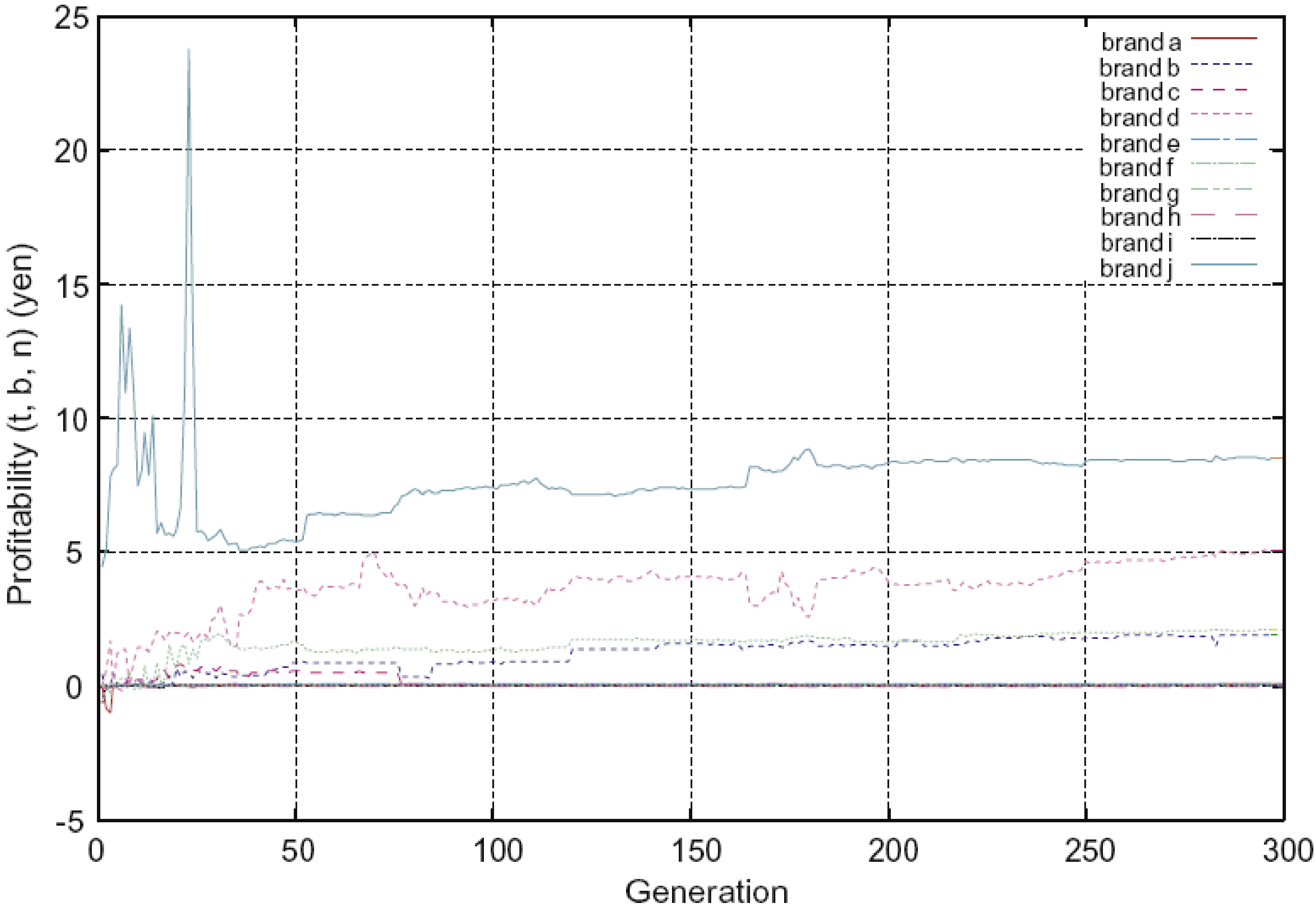


Fig. 9. Profitability change of 10 brands in the training period.

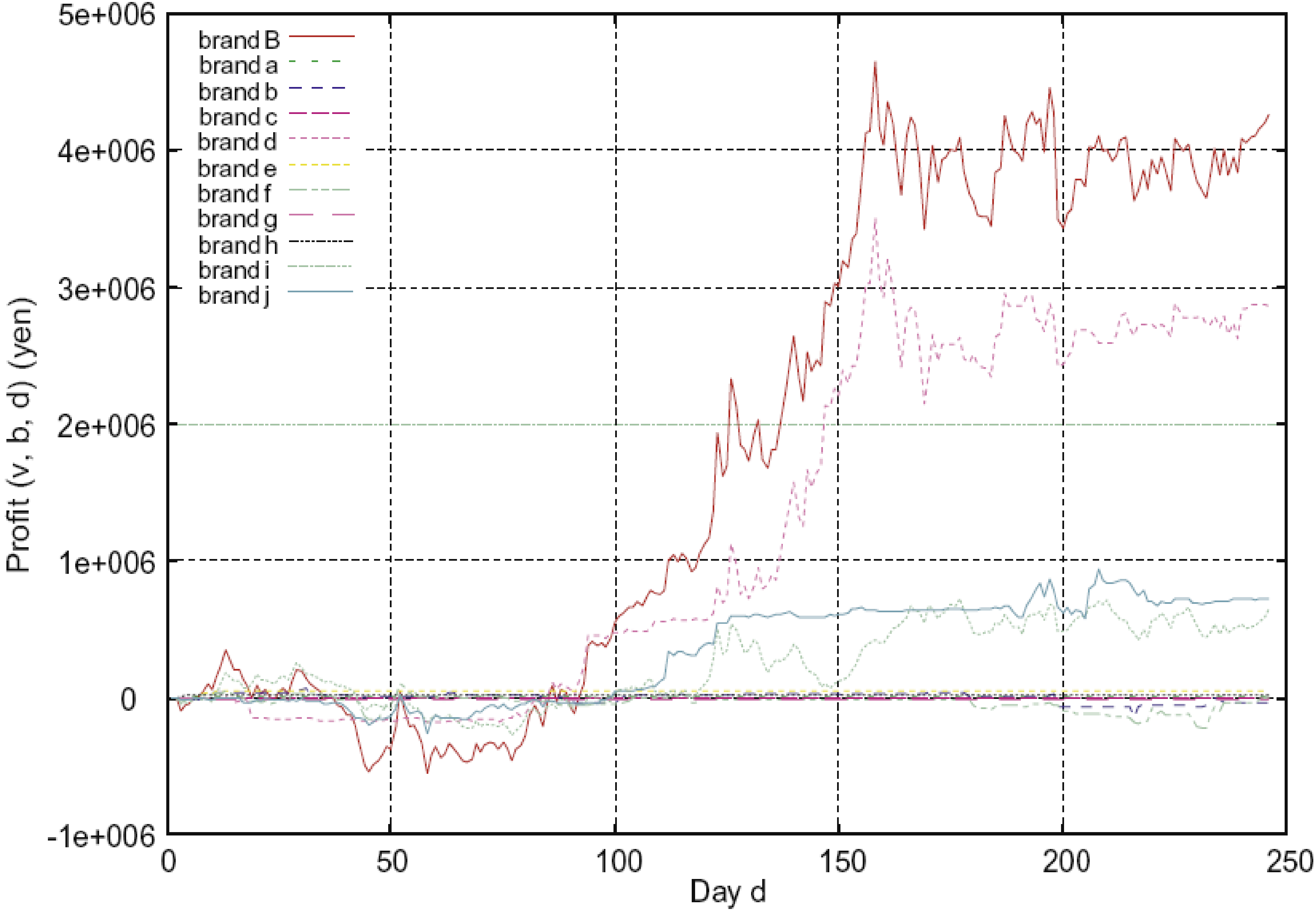


Fig. 10. Profits change of 10 brands in the testing period.

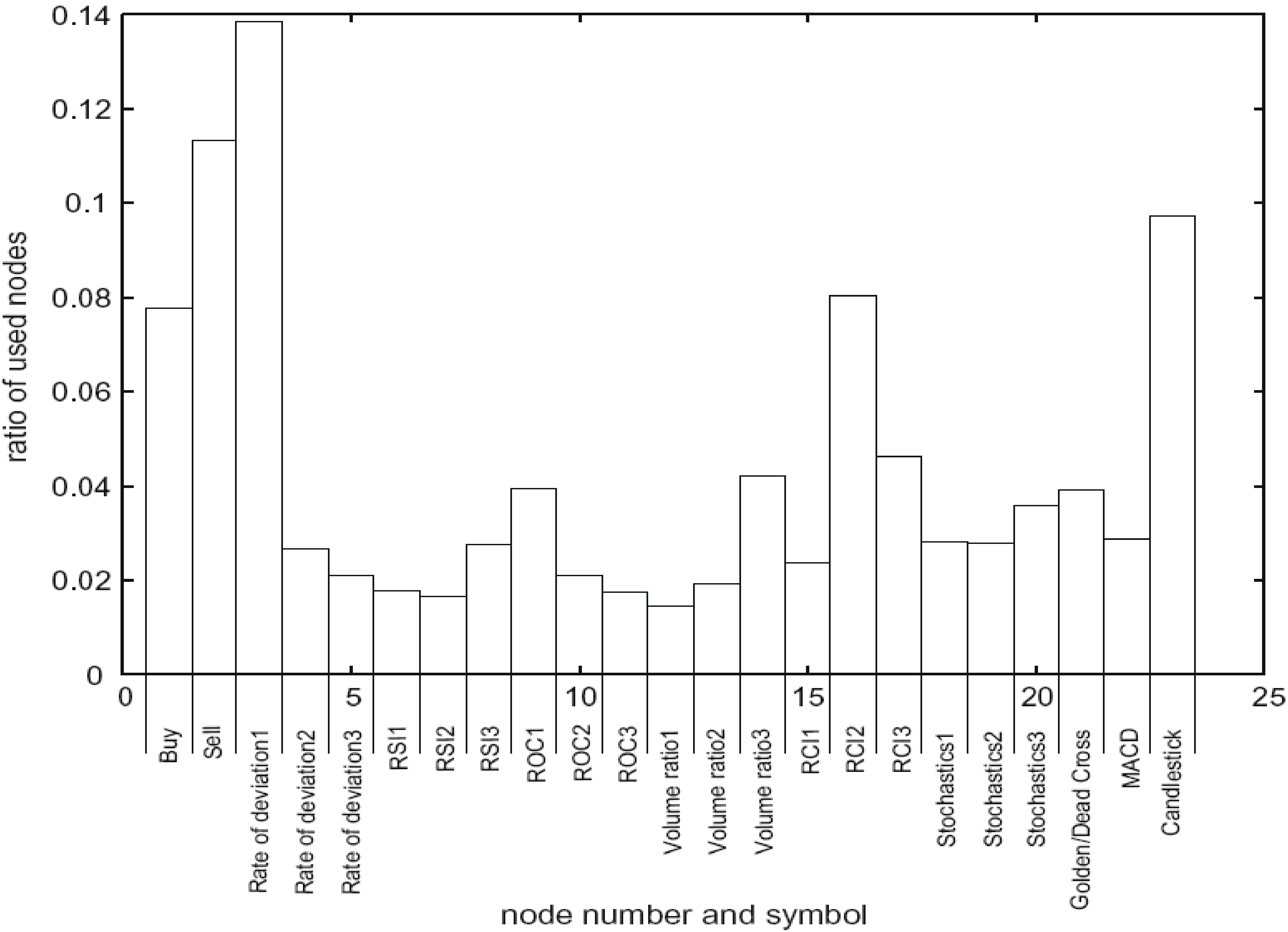


Fig. 11. Ratio of nodes used by GNPcn in the testing period.

Thank you!!!